

RepairDesk

Repair Shop Management System

Product Requirements Document (PRD)

Version 3.0 — March 2026

Last Updated: 25 March 2026

jukibd.lovable.app

1. Executive Summary

RepairDesk is a comprehensive, cloud-based Repair Shop Management System designed for electronics repair businesses (TV, Laptop, AC, Refrigerator, etc.). Built with React 18, TypeScript, Tailwind CSS, and Supabase backend, it provides end-to-end management of repair jobs, customers, financials, inventory, and staff operations.

The system features role-based access control (RBAC) with 19+ permission modules, a public customer portal for job tracking, multi-branch support, SMS notifications, QR code scanning, and a fully responsive design that adapts to any screen size.

2. Technology Stack

Layer	Technology	Purpose
Frontend	React 18 + TypeScript + Vite	SPA with hot reload
UI Framework	Tailwind CSS + shadcn/ui	Design system with dark/light themes
State Management	TanStack React Query	Server state caching & sync
Routing	React Router v6	Client-side navigation
Backend (DB)	Supabase PostgreSQL	Relational database with RLS
Auth	Supabase Auth	Email + Mobile login with OTP lookup
Storage	Supabase Storage	Photo uploads (profile, challan, cheque)
Edge Functions	Supabase Edge Functions	SMS API, customer lookup, password reset
Charts	Recharts	Dashboard analytics visualization
QR Code	html5-qrcode + qrcode.react	QR scan & generate
Export	SheetJS (xlsx)	Excel export for reports
Icons	Lucide React	Consistent icon library

3. Database Schema

The system uses 22+ PostgreSQL tables with Row-Level Security (RLS) policies:

Table	Key Columns	Purpose
jobs	job_number, customer_name, brand_name, model_name, board_name, board_serial, status, service_charge, discount, payable_amount, receive_amount, created_by_name, delivered_by_name	Core repair job records
clients	client_name, contact_number, email, company_name, address, image_url	Customer database
profiles	name, email, mobile, employee_id, nid, address, photo_url, status	Staff user profiles
user_roles	user_id, role (admin/user)	Role-based access control

user_permissions	user_id, module	Granular module permissions
user_activity_logs	user_id, user_name, action, ip_address	Login/logout audit trail
branches	name, phone, email, address, status	Multi-branch locations
brands	name, remarks	Machine brand master data
models	name, remarks	Machine model master data
boards	name, image_url, remarks	PCB board master data
expenses	amount, category_name, expense_date, memo_no, service_provider, tr_no	Expense transactions
expense_categories	name, remarks	Expense categorization
incomes	amount, category_name, income_date, tr_no	Income transactions
income_categories	name, remarks	Income categorization
inventory	part_name, part_number, brand, category, quantity, unit_price, min_stock_level, supplier, location	Spare parts inventory
warranties	job_number, customer_name, warranty_type, start/end_date, terms, status	Warranty tracking
customer_feedback	job_number, customer_name, rating, feedback_text	Customer satisfaction
sms_config	api_key, sender_id	SMS gateway configuration
sms_templates	trigger_status, template_text, is_active	Auto SMS templates
sms_logs	customer_phone, message_text, status, api_response	SMS delivery logs
company_info	company_name, address, phone, mobile, email, logo_url, portal_enabled	Company settings + portal toggle
footer_content	content, is_active	Customizable footer
portal_scroll_messages	message_text, font_size, font_color, is_active	Portal marquee messages
theme_settings	primary_color, sidebar_bg_color, menu_font_size/color, table_font_size/color, page_bg_color	UI theme customization
chart_of_accounts	account_code, account_name, account_type, parent_id, is_active	Accounting chart

4. Authentication & Role-Based Access Control

4.1 Authentication Flow

- Email-based login with Supabase Auth
- Mobile number login via RPC lookup_email_by_mobile function
- Protected routes redirect unauthenticated users to /login
- Auto session refresh with persistent localStorage sessions
- Login/logout events logged to user_activity_logs table

4.2 Role System

Roles are stored in a separate user_roles table (not in profiles) to prevent privilege escalation. The app_role enum supports: admin, user.

4.3 Permission Modules (19 Modules)

#	Module Name	Controls Access To
1	Dashboard	Main dashboard with KPI cards & charts
2	Branch	Branch list, add, edit
3	Machine Data	Brands, Models, Boards management
4	Client Data	Customer list & management
5	Add Job	Create new repair jobs
6	Job List	View all jobs, search, filter, export
7	Collection	Due bill, collection reports
8	Expense / Income	Expense/income categories & entries
9	Ledger	Customer ledger view
10	Reports	Transaction, collection, expense, company-wise, user-wise reports
11	Cashbook	Cash in, cash out, balance sheet
12	Accounts	Chart of accounts
13	Inventory	Spare parts inventory management
14	Warranty	Warranty tracking & management
15	Analytics	Advanced analytics dashboard
16	Feedback	Customer feedback management
17	QR Scanner	QR code scanning for job lookup
18	Tally Export	Tally accounting export
19	Challan Verify	Factory challan verification
20	Setting	Company info, SMS, appearance, footer, notifications
21	Backup Database	Database backup/export

4.4 Admin-Only Features

- Admin menu (User List, Add User, User Logs) visible only to admin role
- Job Delete button visible only to admin users
- Database record deletion restricted to admin
- User management (add, edit, deactivate, permissions)
- Client Portal on/off toggle in Company Info settings

5. Core Modules

5.1 Dashboard

- KPI Cards: Total Jobs, Pending, Completed, Delivered, Total Revenue, Due Amount
- Status-wise pie chart with animated segments

- Monthly revenue bar chart
- Recent jobs table with quick status view
- Fully responsive with card layout on mobile

5.2 Add Job (Multi-Item Support)

Comprehensive job creation form with the following fields:

Section	Fields
Customer Info	Customer Name*, Contact Number, Company Name, Address (auto-fill from existing clients)
Machine Info	Brand Name (dropdown), Model Name (dropdown), Board Name (dropdown), Board Serial Number
Job Details	Job Number (auto-generated), Job Date, Details of Problem, Remarks
Branch	Branch Name (dropdown from branches table)
Financials	Service Charge, Discount, Payable Amount (auto-calculated), Charge Type
Staff	Created By (auto-filled from logged-in user)
Items	Multiple line items with Add Item functionality (item_name, brand_name, model_name, board_name, board_serial, problem_detail)

5.3 Job List

- Searchable by job number, customer name, brand, model, board serial
- Filter by status: All, Received, Diagnosing, In-Progress, Completed, Picked-Up
- Date range filter
- Excel export with SheetJS
- Bulk print selected invoices
- Responsive: table on desktop, card view on mobile
- Pagination with adjustable page size

5.4 Job Detail & Workflow

5-stage linear workflow:

- Received → Diagnosing → In-Progress → Completed → Picked-Up
- Completion Wizard: captures service_charge, discount, payable_amount, charge_type, completed_date
- Delivery Wizard: captures receive_amount, receive_type (Cash/Bank/bKash/Nagad/Rocket), delivered_by_name, delivery_date
- Challan & Cheque image uploads to Supabase Storage
- Admin-only DELETE button for job removal
- Print Invoice with QR code, company branding, amount in words (Bangla)
- SMS notification on status change

5.5 Customer Management

- Full CRUD for client records
- Profile photo upload

- Search by name, phone, company
- Customer job history view
- Auto-fill customer data in Add Job form

5.6 Client Portal (Public)

- Public route at /track (no authentication required)
- Customer enters mobile number to view all their jobs
- Shows company name, logo, and portal scroll message
- Admin can enable/disable portal via Company Info settings (portal_enabled flag)
- Displays job details: job number, brand, model, status, dates, charges
- Full job history with status badges
- Responsive design for mobile access

6. Financial Modules

6.1 Collection

- Due Bill: Lists all jobs with outstanding balance (payable - received > 0)
- Collection Report: Date-wise collection summary with totals
- DUE COLLECTION INVOICE: Printable invoice with QR code

6.2 Expense / Income

- Category management (CRUD) for both expense and income
- Transaction entry with date, amount, category, memo, remarks
- Transaction number (tr_no) tracking
- Service provider details for expenses
- Created-by staff attribution

6.3 Cashbook

- Cash In: All income + job collections
- Cash Out: All expense entries
- Balance Sheet: Running balance with date range filter

6.4 Ledger

- Customer Ledger: Per-customer transaction history
- Shows all jobs, payments, dues for selected customer

6.5 Accounts

- Chart of Accounts: Hierarchical account structure
- Account types: Asset, Liability, Equity, Revenue, Expense
- Parent-child relationships with tree view

7. Reports

Report	Description	Features
Transaction Report	All financial transactions in date range	Date filter, Excel export, admin-only delete
Collection Report	Payment collection summary	Date filter, customer-wise breakdown
Expense Report	Expense summary by category	Date filter, category breakdown
Company-Wise Report	Jobs grouped by company/brand	Brand filter, revenue summary
User-Wise Report	Jobs per staff member	Staff performance metrics

8. Additional Modules

8.1 Inventory Management

- Spare parts tracking: part name, number, brand, category, quantity, unit price
- Min stock level alerts
- Supplier and location tracking
- Search and filter capabilities

8.2 Warranty Management

- Warranty creation linked to jobs
- Warranty types and terms
- Start/end date tracking
- Status management (Active/Expired/Claimed)

8.3 Analytics Dashboard

- Advanced charts and visualizations
- Revenue trends, job completion rates
- Staff performance metrics
- Interactive date range selection

8.4 Customer Feedback

- Star rating system (1-5)
- Text feedback linked to jobs
- Average rating display

8.5 QR Scanner

- Camera-based QR code scanning using html5-qrcode
- Scans job QR codes for quick lookup
- Redirects to job detail page

8.6 Tally Export

- Export financial data in Tally-compatible format

- Excel download for accounting integration

8.7 Challan Verify

- Verify factory challan numbers
- Cross-reference with job records

9. Settings

Setting	Description
Company Info	Company name, address, phone, email, logo upload, portal on/off toggle
SMS Settings	API key, sender ID configuration for SMS gateway
Bulk SMS	Send bulk messages to customers
Portal Scroll Message	Marquee message on customer portal with font size/color
Footer Settings	Customizable footer content with active/inactive toggle
Notifications	Notification preferences
Appearance	Theme customization: colors, font sizes for menu/submenu/table, sidebar/page backgrounds

10. Print & Invoice System

- Customer Receive Copy: Generated on job creation
- Delivery Invoice: Generated on job delivery with full payment details
- DUE COLLECTION INVOICE: For outstanding payments
- All invoices include: Company logo/name/address, QR code, customer details, job details
- Amount in Words: Auto-conversion to Bangla text
- Staff Attribution: Created By and Delivered By names printed
- Authenticated stamp seal on portal-generated prints
- A4 optimized layout with 1-inch margins
- Bulk print support from Job List page

11. Responsive Design

- Fully responsive from mobile (320px) to 4K displays
- Sidebar: collapsible on desktop, drawer overlay on mobile
- Tables: standard table on desktop, card view on mobile
- Header: condensed on mobile with hamburger menu
- Forms: single column on mobile, multi-column on desktop
- Print layouts: A4 optimized regardless of screen size
- Touch-friendly button sizes on mobile
- CSS variables for dynamic font sizing

12. Edge Functions

Function	Purpose	Endpoint
----------	---------	----------

send-sms	Send SMS on job status change	POST /send-sms
send-bulk-sms	Bulk SMS to multiple customers	POST /send-bulk-sms
customer-lookup	Lookup customer by mobile for portal	POST /customer-lookup
reset-password	Admin password reset for users	POST /reset-password
mysql-sync	Legacy MySQL data synchronization	POST /mysql-sync

13. Security

- Row-Level Security (RLS) on all tables
- Roles stored in separate user_roles table (not in profiles)
- has_role() security definer function for RLS policies
- No client-side admin checks (server-validated)
- Session-based authentication with auto-refresh
- Protected routes with ProtectedRoute component
- Admin-only destructive operations (delete)
- Activity logging for audit trail

14. Full Application Prompt

Use this prompt to recreate the complete RepairDesk application:

Build a complete Repair Shop Management System called "RepairDesk" using React 18, TypeScript, Vite, Tailwind CSS, shadcn/ui, and Supabase (Auth, PostgreSQL, Storage, Edge Functions). **AUTHENTICATION:** Email + mobile login (RPC lookup_email_by_mobile), Supabase Auth with session persistence, login/logout activity logging. **ROLES & PERMISSIONS:** user_roles table with app_role enum (admin/user), user_permissions table with 21 modules (Dashboard, Branch, Machine Data, Client Data, Add Job, Job List, Collection, Expense/Income, Ledger, Reports, Cashbook, Accounts, Inventory, Warranty, Analytics, Feedback, QR Scanner, Tally Export, Challan Verify, Setting, Backup Database). Admin sees all + Admin menu. Non-admin sees only permitted modules in sidebar. **SIDEBAR:** Dark gradient sidebar (linear-gradient #0a0f1e to #111936), collapsible on desktop, drawer on mobile. User profile card, nav items with sub-menus, logout button. Icons from Lucide React. **DATABASE:** 22+ tables - jobs, clients, profiles, user_roles, user_permissions, user_activity_logs, branches, brands, models, boards, expenses, expense_categories, incomes, income_categories, inventory, warranties, customer_feedback, sms_config, sms_templates, sms_logs, company_info, footer_content, portal_scroll_messages, theme_settings, chart_of_accounts. All with RLS policies. **CORE PAGES:** Dashboard (KPI cards, pie chart, bar chart, recent jobs), Add Job (multi-item, auto-fill customer, all dropdowns from DB), Job List (search, filter, date range, pagination, Excel export, bulk print), Job Detail (5-stage workflow: Received>Diagnosing>In-Progress>Completed>Picked-Up, completion wizard, delivery wizard, challan/cheque upload, admin-only delete), Edit Job, Customer Management (CRUD, photo, search), Print Invoice (A4, QR code, company brand, amount in words Bangla, staff names). **CLIENT PORTAL:** Public /track route, mobile number search, shows all jobs for customer, company branding, admin toggle (portal_enabled in company_info). **FINANCIALS:** Expense/Income with categories, Cashbook (Cash In/Out/Balance Sheet), Customer Ledger, Collection (Due Bill, Reports), Chart of Accounts (hierarchical). **REPORTS:** Transaction, Collection, Expense, Company-Wise, User-Wise. All with date filters and Excel export. Admin-only delete on transaction report. **OTHER:** Inventory (parts tracking, min stock alerts), Warranty (linked to jobs), Analytics (charts), Feedback (ratings), QR Scanner (html5-qrcode), Tally Export, Challan Verify, Multi-branch support, Bulk SMS. **SETTINGS:** Company Info (logo, portal toggle), SMS Config, Bulk SMS, Portal Scroll Message, Footer, Notifications, Appearance (theme colors, font sizes). **RESPONSIVE:** Auto-adjust all screens. Tables become cards on mobile. Collapsible sidebar. Touch-friendly. Print-optimized A4 invoices. **SECURITY:** RLS on all tables, roles in separate table, has_role() security definer, admin-only delete, activity logging, no client-side admin checks.